Final Project | Connor Oman

Introduction:

My name is Connor Oman and I'm a graduate student here at the University of Indiana. My ultimate goal in completing this masters degree in data science from the University of Indiana is to land a data science job within the gaming industry. Therefore, the most logical choice was to narrow my focus for this project on data from the gaming industry. Specifically, I chose to source data from Steam (the premier distribution platform for PC gaming) because it's publicly available and provides relevant information for both indie and mainstream studios to make insightful decisions. Steam allows companies to list the games with a large assortment of tags and genres to further advertise their game. Narrowing the scope even further, I wanted to see how the presence of these tags and genres impacted the likelihood of a user purchasing the game or leaving a positive review. The overall quality of a game drives an individual purchase, but analyzing the prevalence of tags and genres in games that are highly reviewed or purchased would help us understand what's currently popular amongst users. Going further, I plan to also compare prevalence to yearly market volume to quantify oversaturation in specific tags and genres. For this project, I've created two models that predict the review_ratio and estimated_owners respectively from numeric and categorical columns, found the genres and tags most common in well performing games, compared those genres and tags to the saturation in the market, and explored how price and average playtime affect performance.

Methodology:

The dataset used in this project is sourced directly from Steam itself. To accomplish this I followed steps provided by Kaggle users and guidance from some of my close friends. For the scope of this project, I included data from 2003 (generally the year Valve launched Steam) to March 2025 or approximately 1 year ago. Therefore, the dataset includes information for every game released on steam between 2003 and 2025 including the release date, a range of estimated owners, the number of positive & negative recommendations, lists of genres & tags, and tons of additional information.

Cleaning the data went smoothly as most columns presented valid data, but there were still some issues with balance and skew. Many games are listed on steam with very little interaction from consumers. It’s not uncommon for a game to be listed with zero downloads or have no playtime. Therefore, I decided to only analyze games that at the very least have an average_playtime_forever score recorded. This massively cut down the size of the dataset from 90k rows down to 8000. This indicates that most listings on steam go completely unknown which also highlights the competitiveness of steam. However, I had believed that 8000 rows was still enough for models and machine learning to be accurate enough to provide credible insight.

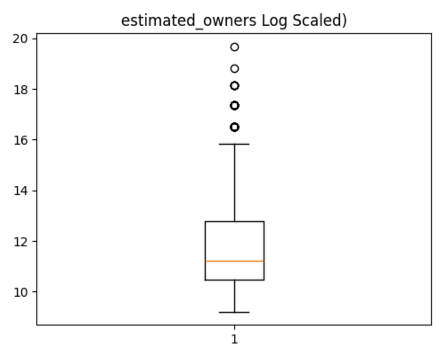
For quantifying the overall public opinion on a game, the Metacritic score can be useful, but as a platform, it’s often vulnerable to brigading as users don’t necessarily have to buy the game to leave a review. Thus, I wanted a better, more accurate source to quantify review score. On steam, you can only recommend or not recommend a game after you buy it. Recommending a game counts as a positive review and not recommending a game counts as a negative review. I took the positive column and divided it by the negative and positive columns added together to find the ratio of positive reviews. Likewise, direct sales information for each game isn’t public by default. However, steam does provide an estimated_owners which is a range of possible owners like 100 - 2000 which provides a general understanding of how large a game is. This range was technically a string object and I had to convert the column to a numeric measurement. I decided to take the average between the lower and upper bounds to get a rough estimate of the number of owners.

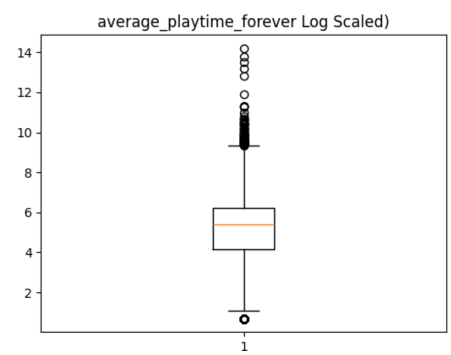
The very first step was to take descriptive statistics of the key 8 most relevant columns I was using for my analysis. These statistics helped establish the context and direction for the rest of my project.

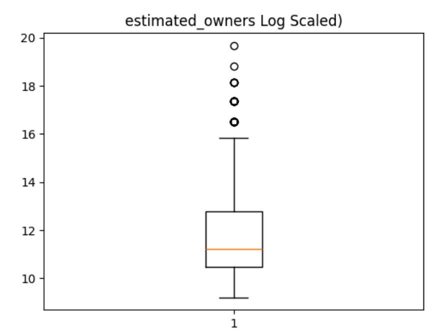
	positive	negative	estimated_owners	average_playtime_forever	median_playtime_forever	metacritic_score	price	review_ratio
count	8.019000e+03	8.019000e+03	8.019000e+03	8.019000e+03	8.019000e+03	8019.000000	8019.000000	7826.000000
mean	9.031433e+03	1.449033e+03	6.200293e+05	1.285980e+03	1.283100e+03	15.466517	11.848545	0.780799
std	9.818226e+04	1.622574e+04	5.050763e+06	2.275021e+04	2.941712e+04	30.703869	12.385913	0.155336
min	0.000000e+00	0.000000e+00	1.000000e+04	1.000000e+00	1.000000e+00	0.000000	0.000000	0.000000
25%	1.640000e+02	4.500000e+01	3.500000e+04	6.400000e+01	6.000000e+01	0.000000	1.990000	0.699373
50%	6.390000e+02	1.390000e+02	7.500000e+04	2.180000e+02	2.130000e+02	0.000000	8.990000	0.816219
75%	2.602500e+03	4.690000e+02	3.500000e+05	5.110000e+02	4.430000e+02	0.000000	18.990000	0.899605
90%	1.082000e+04	1.768200e+03	7.500000e+05	1.436200e+03	1.108600e+03	77.000000	29.990000	0.945081
max	7.480813e+06	1.135108e+06	3.500000e+08	1.462997e+06	1.462997e+06	97.000000	99.990000	1.000000

The steam review scores indicate that most games receive an approval rate of roughly 0.78%. The metacritic score appears rather low, but that can be easily explained by many games having review data on Steam, but not on Metacritic as shown by the 0th to 75th percentile being at 0.0 out of 100. Average playtime is interesting because the standard deviation is massive all while the max is 3 magnitudes greater than the 90th percentile. The column shows that most games on steam are played briefly compared to fewer games being played a ton. Similarly, the price indicates most games are cheap with the mean and median landing under 12 dollars. This price mark is likely due to the majority of games listed on Steam being from smaller scoped indie games as opposed to big-studio releases that usually center around 60 dollars.

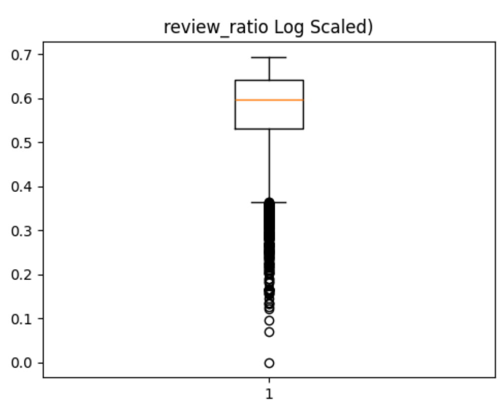
Next, I looked at the presence of outliers within the dataset. It's very clear from the numeric overview that most columns have a few extremely large measurements which correspond to massive game titles. The log scaled box plots of many columns further support this showing the majority of outliers occurring past the upper bound.





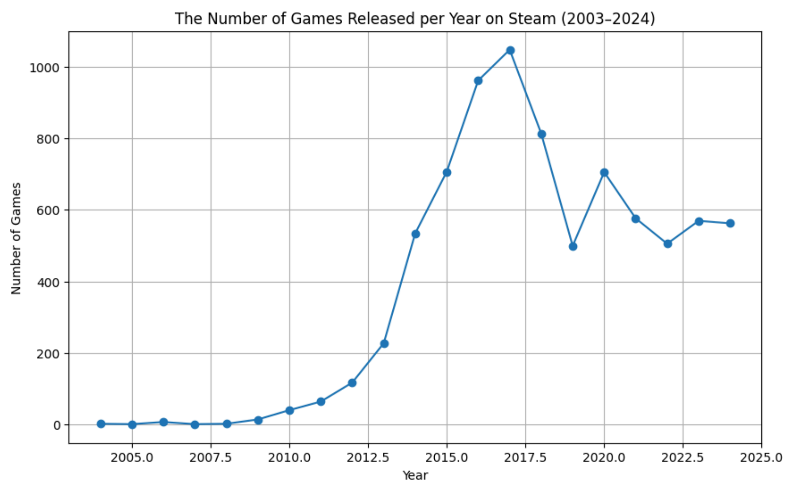


However, there was a single exception. The only column that had the vast majority of outliers below the lower bound was review_ratio. This confirms what I already believed that any

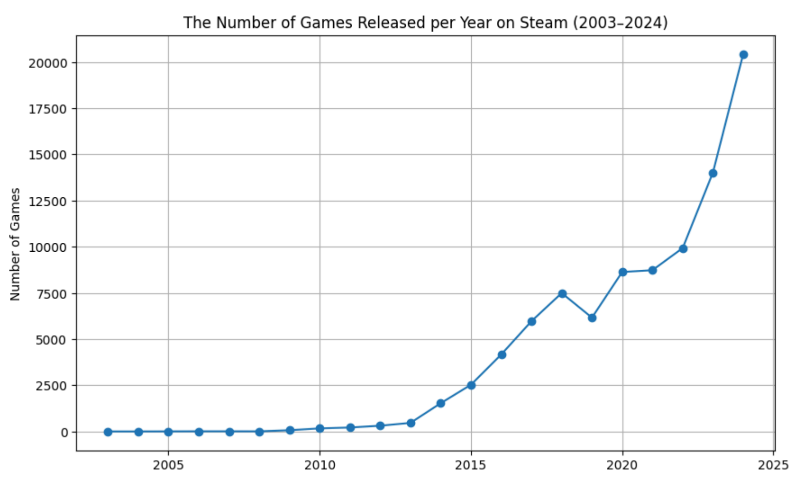


approval rate below 40% is basically an anomaly. Given the competitiveness of steam, users have a wide range of choices and are less likely to buy a product they think they won't enjoy. I designed a cell to remove outliers past a specified range to promote the accuracy of models later in the project. However, I chose not to use it for predicting review ratio as games could only score within range of 0.0 to 1.0. I used it sparingly for predicting estimated owners, but I didn't exclude the outliers when calculating the prevalence of the best tags and genres because the outliers had the same amount of impact when counting tag/genre occurrence as any other game.

To further establish context, I wanted to see how the number of games released per year has changed over time and create a release_year column that would become relevant later in analysis. For the sake of the graph, I excluded 2025 as it only had 3 months of measurement instead of 12. The all time peak can be observed in 2016 and the graph rapidly picks up pace within the 2012 to 2014 time period. After 2016, the graph appears to stabilize with games released oscillating between 450 and 650 games.



This graph only shows the games included in the 8000 game subset with recorded playtime. To see how the trendline compares to all games being released on Steam, I made a second graph that includes all of the data before it was subsetted for average_playtime_forever > 0.



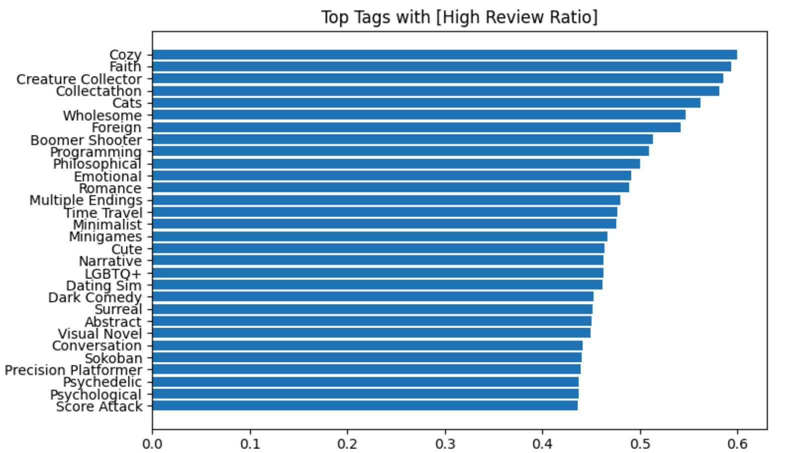
This graph shows a consistent exponential increase in games from 2003 to 2024. The two graphs show that the overall volume of games being released each year has consistently increased over time. However, the overall popularity of those games has not increased proportionally as a vast majority of those games aren't included in the subsetted dataset.

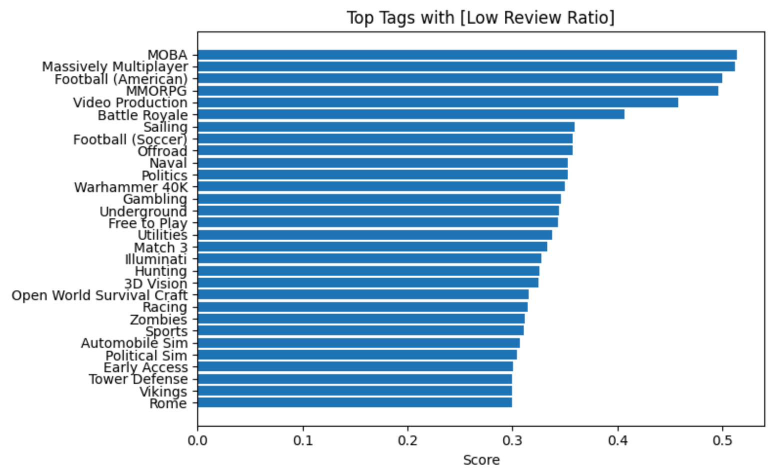
Results:

After the data was cleaned and the context was established, I started the analysis by creating 2 machine learning regression models to predict the review ratio and estimated owners. For both models, I used 3 numerical columns consisting of average_playtime_forever, metacritic_score, and price. I also used 2 categorical columns tags and genre which will become extremely relevant later in the project. The first model (where I attempted to predict the review_ratio) performed alright. It received a R-squared value of 0.3 indicating 30% of the variance was explained by the model whereas the RSME was relatively low at 0.018. However, the cross validation R-squared scores were wildly inconsistent and the average was actually below the mean. Therefore, I'm inclined to believe the model doesn't generalize enough and is likely sensitive to the split. Similarly, the second model predicting estimated_owners performed worse. Given that the data was skewed I performed a log transformation. The R-squared value was 0.42, but all four cross validation scores were below the mean and RSME was above 1. Out of curiosity I calculated the impact the numerical columns had on the predictions and they had very minimal impact at roughly 3% & 10% respectively. Unfortunately, the dataset does not perform well under machine learning models which suggests the root cause behind review_ratio and estimated_owners to be outside the dataset.

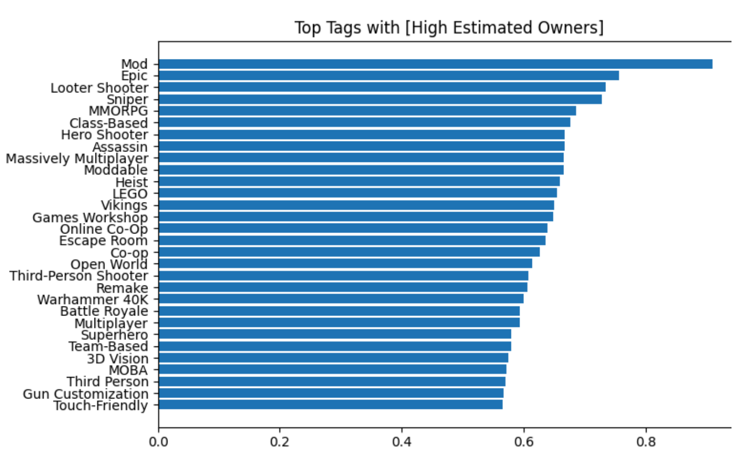
However, I believe there is still valuable insight to be gained from the tags and genres listed on Steam. I designed two functions that could count the prevalence of each tag and genre. In get_top_relative_terms() I made sure to calculate relative prevalence rather than overall prevalence because some terms like "multiplayer" are more common than those like "cozy". Therefore, it's more insightful to compare each term's performance to their overall prevalence in the dataset. Then using quantile(), I could define those thresholds and calculate the prevalence of specific tags and genres in the top 25% and bottom 25% of reviewed games. In doing the same for estimated_owners while keeping them separate, I created multiple top 30 lists that display the most common tags & genres according to review_ratio & estimated_owners.

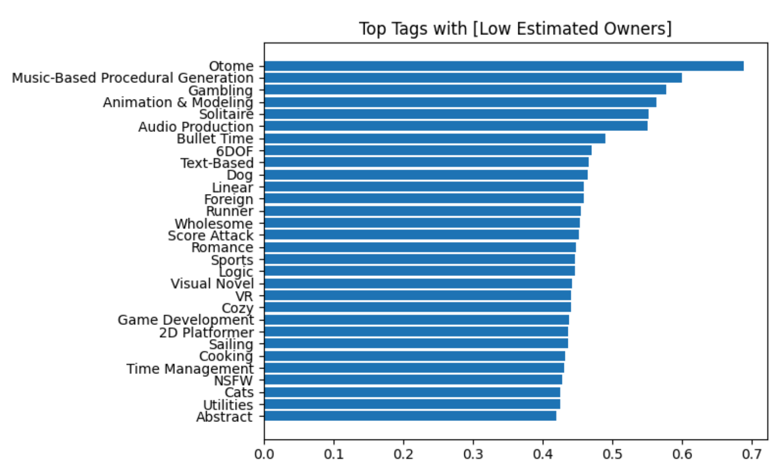
I generated 8 bar charts that showed the proportion of games with that specific tag in the top 25% of reviewed and owned games. As an example, you can see the best and worst performing tags according to review ratio below:





The top performing tag was cozy with approximately 60% of all games tagged with cozy landing in the top 25% of all games reviewed on steam. The worst performing tag was MOBA which stands for Multiplayer Online Battle Arena which isn't surprising as most MOBAs have a reputation of being highly competitive. Generally, more atmospheric & narrative adjacent tags related to establishing the theme & feeling of the game (like Cozy, Wholesome, Faith, Philosophical, etc) tend to score better as they are relatively uncontroversial. Whereas, the tags that performed poorly generally fall into two camps. First being highly competitive genres like MOBAs or MMORPGs where the standard is set extremely high. Second being a disconnect between the target audience. Some tags like both footballs, sports, gambling, and match 3 are more popular on other platforms like console and mobile. However, just because a game has a great review score doesn't automatically mean it has a proportional amount of owners. Thus, I found the best and worst tags according to estimated_owners:

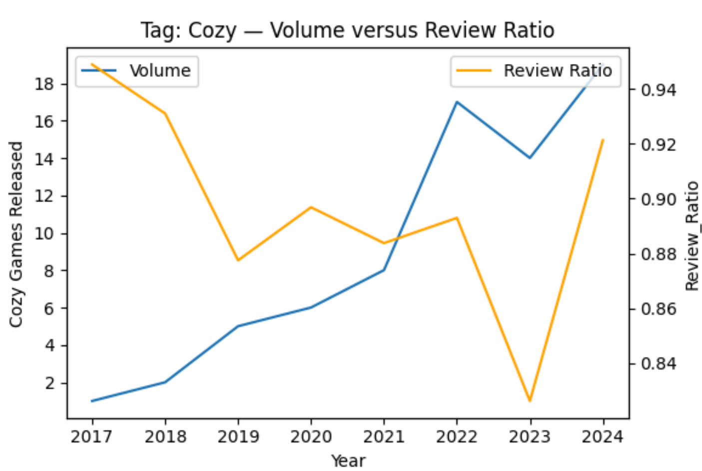




Mod was the best performing tag whereas Otome was the worst according to estimated_owners. Mod scored well because mods are generally free and according to analysis later in the project, it was relatively uncommon with multiple incredible performances. Generally, games that require large communities of people to function (like MMORPG, MOBA, Team-Based, Massively Multiplayer, etc) performed better in terms of estimated_owners. Technically, you can also list software on Steam instead of your typical video game. Thus, many poorly performing tags (like Animation & Modeling, Audio Production, Game Development, etc) fell into that archetype. Interestingly, many tags that performed well in terms of review_ratio performed poorly in estimated_owners which further supports the dichotomy between review_ratio and estimated_owners. The recipe for making a well reviewed game is not the same for a well owned game and companies should ensure they are aligning their games with their priorities.

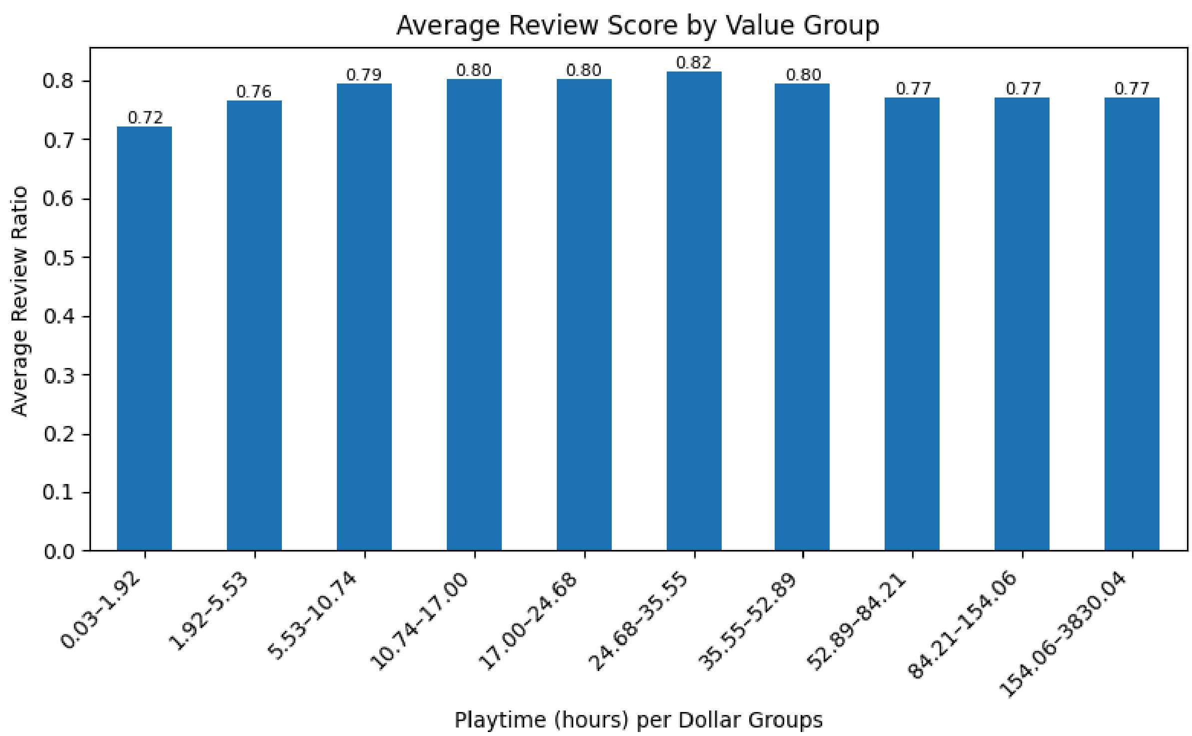
Like I did for tags, I made another four charts that ranked the best and worst genres according to review_ratio and estimated_owners. However, the trends in those charts were largely the same and in depth analysis goes past the scope of this report. If you'd like to see the genre rankings, you can load up the .ipynb or easily view them through the provided example .html on GitHub.

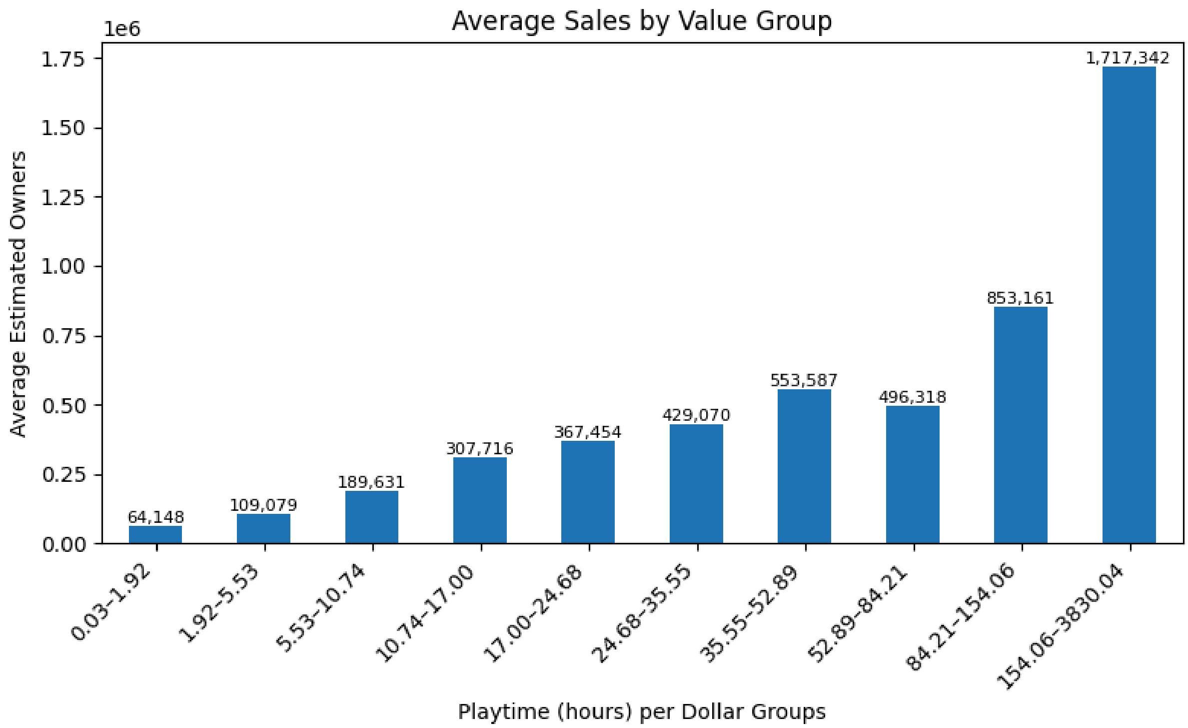
Going further, I was curious if specific tags/genres were becoming oversaturated in the market. While tracking a term's performance over the entire 20 year time span is helpful, understanding the term's projected performance in the current market is arguably more insightful. Therefore, for every tag & genre on the previous rankings, I calculated the total volume of those tags being released on steam each year. Then I used .corr() to see how review_ratio and estimated_owners changed in response to an increase in volume. Finally, I created a dynamic plotting system that lets you input a tag or genre and plots the average review_ratio or estimated owners over time compared to the volume. Let's use Cozy as an example. Review_ratio was negatively correlated to volume with -0.441 and estimated_owners was also negatively correlated



to volume with -0.369. This is further supported by the plot which roughly shows an inverse relationship between review_ratio and volume. However, the scale of volume is rather small which indicates that randomness could be a causation. Still, the vast majority of tags & genres saw a negative correlation between volume and review_ratio/estimated_owners with the exception of a few tags like faith. Ultimately, the best performing tags with saturation correlation above -0.01 were Hero Shooter, Vikings, Escape Room, Open World, and Third-Person Shooter. The top 3 genres were Racing, Utilities, and Design & Illustration.

Lastly, I wanted to see how the playtime to dollar ratio of a game impacted price. I took the total playtime in hours and divided it by the price to get a rough estimate of the ratio. In dividing the ratios into 10 bins, I created two visualizations for average review_ratio and average estimated_owners as seen below:





The average review_ratio remained relatively consistent throughout every bin with the exception of 0.03 to 1.92 which was 0.05 points lower than any other bin. However, despite each bin having an equal number of games, the average estimated_owners chart is clearly skewed towards the right. There is a clear linear increase in estimated users until the final bin. This intuitively confirms that the games receiving the most playtime are also extremely popular. Review_ratio appears to be largely independent from playtime compared to estimated_owners.

Conclusion:

Throughout this project, deriving consistent predictive insights from the dataset proved challenging. The models appeared to perform adequately at first, but the cross validation proved that the models were susceptible to the random split. Whether this was caused by the subset of 8000 games or skew is largely irrelevant because there are still numerous external factors outside of the dataset that influence review_ratio & estimated_owners. However, by pivoting to counting tags and genres directly and their relative prevalence, I gained insightful information about the best tags & genres. More importantly I learned different tags like Cozy were more likely to contribute to a high review score than total owners compared to other tags like MMORPG. I expanded upon this topic by quantifying the market saturation. Most tags & genres negatively corresponded to increases in volume, but I identified key terms that were popular and comparatively undersaturated. I visualized the impact of playtime to price ratio and learned that it has very little impact on the review_ratio, but a significant impact on the estimated_owners.

Ultimately, although the predictive modeling component did not perform as consistently, I believe I gave an honest assessment of the shortcomings, derived useful insights from tag & genre prevalence, and gained valuable experience. Subsetting the dataset down to 8000 rows could've provided some potential bias. However, with my original vision for the project, I was unsure how to incorporate the other 82000 rows. Thus, for future analysis, I think it would be interesting to incorporate every row and try to identify what makes a game unknown or undiscoverable. Overall, I had fun with this project and it makes me excited to do similar analysis in the future.